

# DecisionIQ Master Handbook

## CHAPTER 13: The Insight Engine, Prescriptive Guidance, and C-Suite Visualization

### 13.1 The Insight Engine

The core differentiator of DecisionIQ is the **InsightEngine**—a Python diagnostic layer that correlates multi-department database logs to automate root-cause auditing:



- Revenue Drop Detection:** Checks if Net Sales Revenue dropped.
- Refund Correlation:** Correlates the drop with the 18% refund spike in North America during Q2 2025.
- Fulfillment Link:** Connects refunds to the 55% late delivery rate at warehouse **WH002**.
- Supplier delay Link:** Correlates warehouse delays with component shipping delays from supplier Apex Technology (**SUP001**).
- Ticket Link:** Traces complaints to the logistics ticket storm and CSAT crash to 1.8.

### 13.2 Prescriptive Recommendation Algorithms

Every diagnostic alert produces actionable recommendations:

- Action:* Shift 40% of hardware orders to EuroChip AG (dual-sourcing). (Est. value: \$180,000/year).
- Action:* Reallocate 15% safety stock to warehouse **WH002**. (Est. value: \$120,000/month).
- Action:* Trigger customer success outreach for accounts that experienced late shipments. (Est. ARR preserved: \$150,000).

### 13.3 Executive UI Dashboard Design

The dashboard is styled using professional consulting standards:

- Palette:** Navy Blue (**#1B2A4A**), Accent Gold (**#C5A059**), Cool Slate (**#F4F5F7**), and Soft Crimson (**#C62828**).
- Typography:** Google Inter and Outfit fonts.
- Visual Layouts (Streamlit):** Renders role-based tabs:
  - CEO Overview:* Top-level KPIs, revenue forecasts, and the automated Insight Feed.
  - Finance:* Waterfall charts reconciling gross bookings to net sales.
  - Operations:* Warehouse late shipment bars and supplier reliability scatter plots.
  - Customer Success:* Cohort retention heatmaps and ML-predicted churn risks.
  - Marketing:* ROAS indicators and ad campaign efficiency scorecards.